A V-ADMISSIBLE BLOWUP MAKES THE STRICT TRANSFORM AN ISOMORPHISM ALONG THE CLOSURE OF T

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3296-3305 of sources/stacks-spaces-pushouts/pinned-original-source.tex; proof span is lines 3307-3337, the complete author proof (29 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 17 of queue_spaces-pushouts.json (pb=1617, stmt 3296-3305, proof 3307-3337).

Editorial note. spaces-pushouts wave-1: fresh adjudication accept (2026-10-14); six-check dispatched pool (§2 row appended 02:4x).

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-blowup-iso-along' also exists in stacks-flat — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0F46: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3240–3252. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 086D: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3337–3337. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 087F: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3337–3337. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0A4G: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3337–3337. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:spaces-pushouts.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a proper morphism of quasi-compact and quasi-separated algebraic spaces over S. Let $V \subset Y$ be a quasi-compact open and $U = f^{-1}(V)$. Let $T \subset |V|$ be a closed subset such that $f|_U : U \to V$ is an isomorphism over an open neighbourhood of T in V. Then there exists a V-admissible blowing up $Y' \to Y$ such that the strict transform $f' : X' \to Y'$ of f is an isomorphism over an open neighbourhood of the closure of T in $|Y'|$.*

Proof. Let $T' \subset |V|$ be the complement of the maximal open over which $f|_U$ is an isomorphism. Then T', T are closed in $|V|$ and $T \cap T' = \emptyset$. Since $|V|$ is a spectral topological space (Properties of Spaces, Lemma 0A4G) we can find constructible closed subsets T_c, T'_c of $|V|$ with $T \subset T_c$, $T' \subset T'_c$ such that $T_c \cap T'_c = \emptyset$ (choose a quasi-compact open W of $|V|$ containing T' not meeting T and set $T_c = |V| \setminus W$, then choose a quasi-compact open W' of $|V|$ containing T_c not meeting T' and set $T'_c = |V| \setminus W'$). By Lemma 0F46 we may, after replacing Y by a V-admissible blowing up, assume that T_c and T'_c have disjoint closures in $|Y|$. Let Y_0 be the open subspace of Y corresponding to the open $|Y| \setminus \overline{T}'_c$ and set $V_0 = V \cap Y_0$, $U_0 = U \times_V V_0$, and $X_0 = X \times_Y Y_0$. Since $U_0 \to V_0$ is an isomorphism, we can find a V_0-admissible blowing up $Y'_0 \to Y_0$ such that the strict transform X'_0 of X_0 maps isomorphically to Y'_0, see More on Morphisms of Spaces, Lemma 087F. By Divisors on Spaces, Lemma 086D there exists a V-admissible blow up $Y' \to Y$ whose restriction to Y_0 is $Y'_0 \to Y_0$. If $f' : X' \to Y'$ denotes the strict transform of f, then we see what we want is true because f' restricts to an isomorphism over Y'_0. □